Numeri Casuali

Calcolatore genera numeri pseudo-casuali con un algoritmo

$$(5x+3) \% 16 = 6$$

$x = 7 \Rightarrow \frac{38}{16} = \frac{32+6}{16} = 2\frac{6}{16}$

$x = 6 \quad \underline{1}$

$\Rightarrow [0, \dots, 15]$

si ripetono. Solo 16 # distinti;

$(ax+b) \% c$ algoritmo di base.

```
#include <stdlib.h>

int main() {
    =
    double x = drand48();
```

drand48(): genere # casuali in $[0,1)$

2^{48} periodo

seme.c

```c
#include <stdio.h>
#include <stdlib.h>

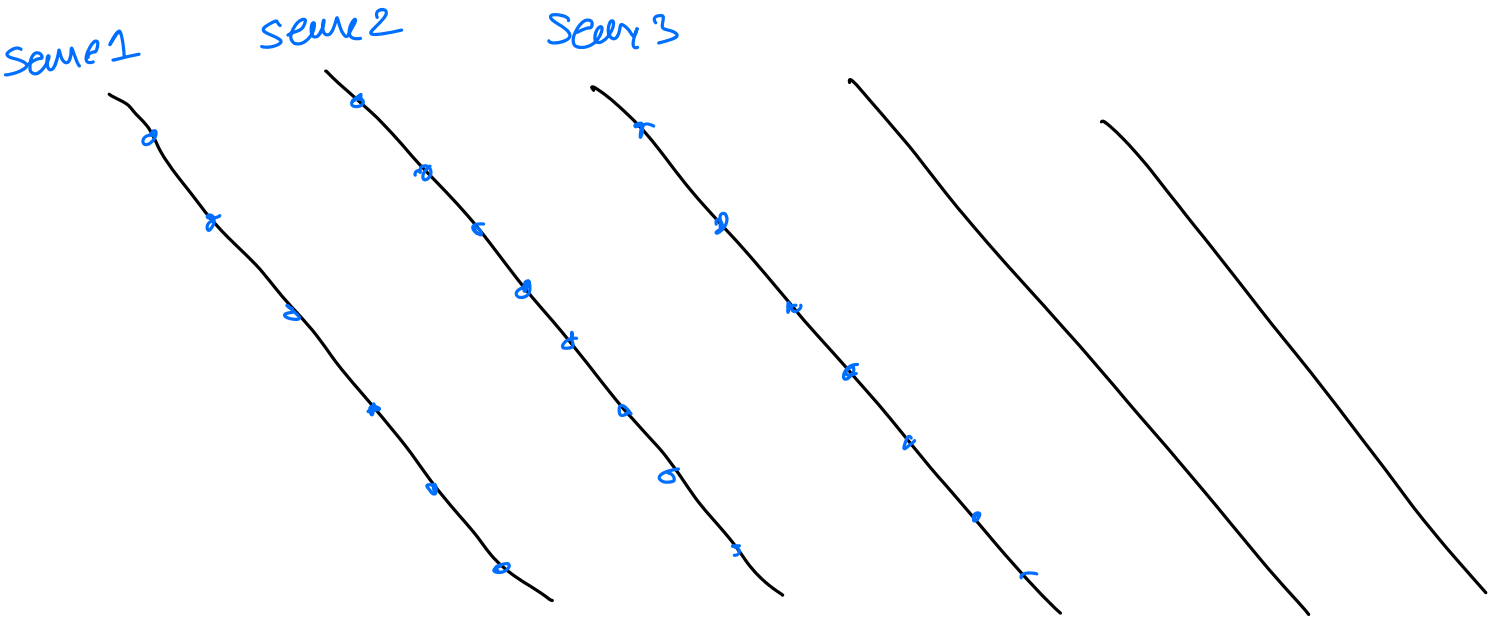
int main() {
    long seme;
    int i;
    printf("seme: ");
    scanf("%ld", &seme);
    srand48(seme);
    for (i = 0; i < 3; i++) {
        printf("%.4f\n", drand48());
    }
}
```

genera

Terminale: seme.c

```
$ gcc -Wall -Wextra -o app.exe seme.c -lm
$ ./app.exe
seme: 5
0.5248
0.2729
0.7886
$ ./app.exe
seme: 5
0.5248
0.2729
0.7886
$ ./app.exe
seme: 6
0.3956
0.9774
0.5270
```

inizializza il seme della sequenza pseudo-cas.



srand48(): 1 sola volta per tutto il programma

drand48(): ogni volta che ci serve un numero cas.

Come sceglie seme a caso ? Seme ≡ long

time(NULL) equivale a time(0)

srand48(time(0)) time() in <time.h>

time(0): tempo in sec. dal 1970-01-01 00:00

srand48(): inizializza seme (1 sola volta)

drand48(): genera $x \in [0,1)$ # razionali

Generare interi:

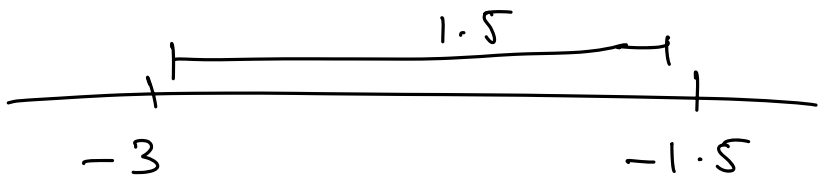
lrand48(): $n \in [0, RAND_MAX]$ interi

Generazione di $x \in [a,b]$

$x \in [0,1) \longrightarrow y \in [a,b]$

$y = a + (b-a)x \qquad b > a$

$y \in [-3, -1.5] \qquad y = -3 + 1.5 * x$



$y = a + (b-a) * drand48()$

$y \in [a,b) \qquad b > a$

angolo.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

int main() {
    double a = 10., b = 80.;   // in gradi
    double angolo;
    int i;
    srand48(time(NULL));
    for (i = 0; i < 6; i++) {
        angolo = a + (b - a) * drand48();
        printf("%.2f\n", angolo);
    }
}
```

Terminale: angolo.c

```
$ gcc -Wall -Wextra -o app.exe angolo.c -lm
$ ./app.exe
59.26
33.94
63.99
25.44
48.49
38.05
```

Dado a 6 facce

1, 2, 3, 4, 5, 6 risultati
a = 1, b = 6

int y = a + (b-a) * drand48()

$y \in [1, 6)$

1.000 -
:
5.999 - -

$\Rightarrow y \in [1, 5]$ interi

b=6 a=1

y = (b-a+1) * drand() + 1

$y \in [0, 6] \longrightarrow [1, 7)$

[0, 1, -- , 5]

1.00 - -
:
6.99 —

int y = (b-a+1) * drand48() + 1

$\in [1, --, 6]$

dado uscudo drand48()

Dado con lrand48() uscudo %

b=6
a=1

lrand48() $\in [0, RAND_MAX]$ interi

lrand48() % (b-a+1) $\in [0,1,2,3,4,5]$

```
int dado = lrand48()%(b-a+1) + 1
```

intervallo $[a,b)$ reale

intervallo $[a,b]$ intero

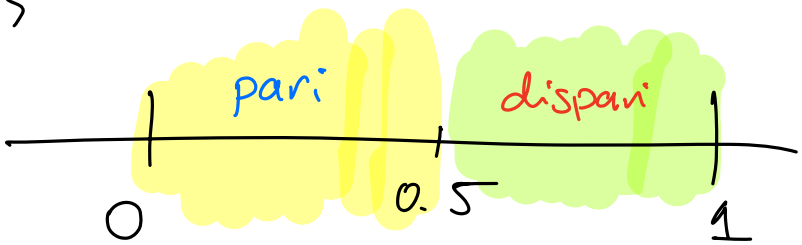
pari / dispari

lrand48() % 2 $\in [0,1]$

non ottimale

```
x = drand48()   ∈ [0,1)

if (x < 0.5) {
    printf("pari\n");
} else {
    printf("dispari\n");
}
```



int y = 2*drand48()

y = 0 , 1

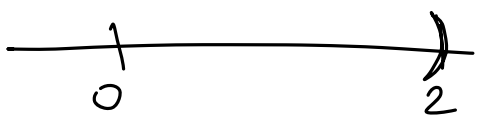
if (y == 0) ?

"pari"

? else ?

"dispari"

?



Distribuzione uniforme

uniformi.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
#define N 100000

int main() {
    FILE *fp;
    int i;
    srand48(time(NULL));
    fp = fopen("casuali.txt", "w");
    if (fp == NULL) {
        printf("errore: impossibile aprire casuali.txt\n");
        exit(-1);
    }
    for (i = 0; i < N; i++) {
        fprintf(fp, "%.8f\n", drand48());
    }
    fclose(fp);
}
```

Terminale: uniformi.c

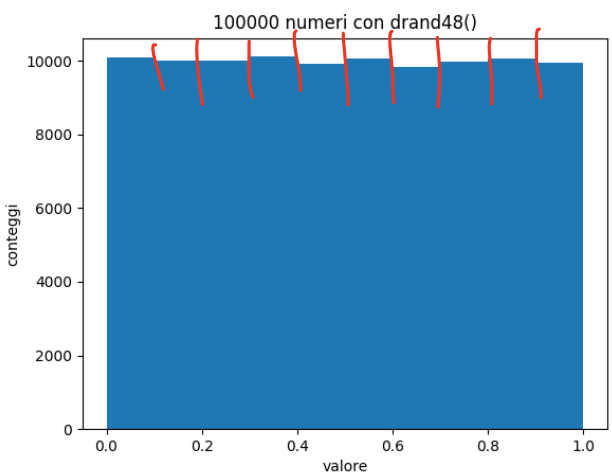
```
$ gcc -Wall -Wextra -o app.exe uniformi.c -lm
$ ./app.exe
$ wc -l casuali.txt
  100000 casuali.txt
$ head -3 casuali.txt
0.70365322
0.34193120
0.77129229
```

istogramma.py

```python
import numpy as np
import matplotlib.pyplot as plt

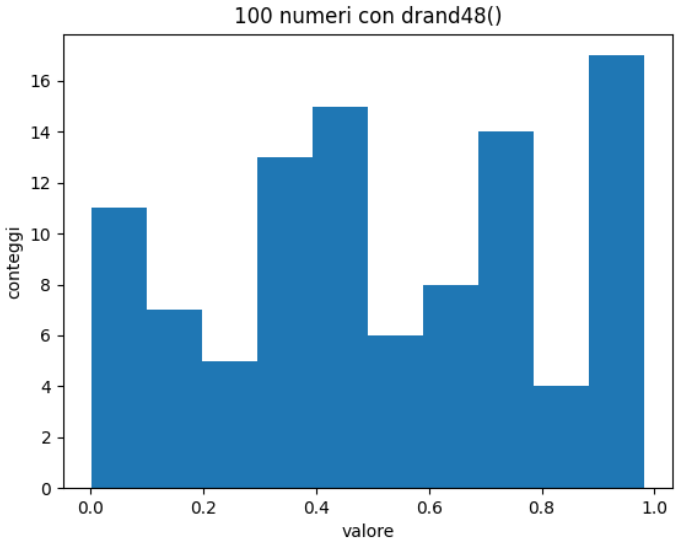
x = np.loadtxt('casuali.txt')   # uno per riga

plt.hist(x, bins=10)
plt.xlabel('valore')
plt.ylabel('conteggi')
plt.title('100000 numeri con drand48()')
plt.savefig('istogramma.png')
plt.show()
```

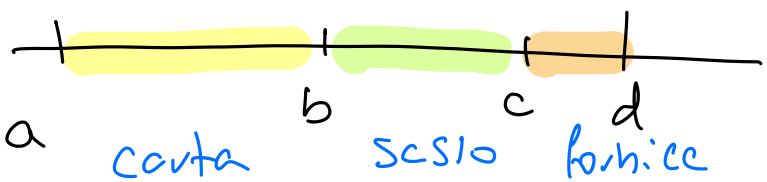


bin ≡ intervallo.

$X \in [0,1)$ intervallo largo $\frac{(1-0)}{\#bin} = 0.1$



Distrib. non uniforme



$x \in [a, d]$ $\quad x = a + (d-a) * drand48();$

```
if ( x < b ) {
    printf (" carta \n");
} else if ( x < c && x > b ) {
    printf (" sasso \n");
} else {
    printf (" forbice \n");
}
```

inutile

Distrib. non uniforme

voglio generare testa con prob. di p% p = 70%

```
if ( drand48() < 0.7 ) {
    printf(" testa \n");
} else {
    printf (" croce \n");
}
```